0	1	2	3	4	5	6	7	8	9	10	11	12
1	1	2	3	4	5	6	7	8	9	10	11	12
2	2	4	6	8	10	12	14	16	18	20	22	24
3	3	6	9	12	15	18	21	24	27	30	33	36
4	4	8	12	16	20	24	28	32	36	40	44	48
5	5	10	15	20	25	30	35	40	45	50	55	60
6	6	12	18	24	30	36	42	48	54	60	66	72
7	7	14	21	28	35	42	49	56	63	70	77	84
8	8	16	24	32	40	48	56	64	72	80	88	96
9	9	18	27	36	45	54	63	72	81	90	99	108
10	10	20	30	40	50	60	70	80	90	100	110	120
11	11	22	33	44	55	66	77	88	99	110	121	132
12	12	24	36	48	60	72	84	96	108	120	132	144

LCM Least Common Multiple no "greatest common multiple".
Prime factorisation of 2 to 10:
$2=2^1$, $3=3^1$, $4=2^2$, $5=5^1$, $6=2^1x3^1$, $7=7^1$,
$8=2^3$, $9=3^2$, $10=2^1x5^1$
LCM(2 to 10)$=2^3x3^2x5^1x7^1 = 2520$ (select highest powers)

Primes (Product of Primes = Prime Factorisation)
Primes 1 to 100: 2,3,5,7,11,13,17,19,23,29
,31,37,41,43,47,53,59,61,67,71,73,79,83,89,97
If you see (mod n): often multiples/LCM — looking for x that is a multiple of n.
If you see f(n)=0: factors/roots — finding N to make equation zero (e.g., Rational Root Theorem to find possible root(s)).

Log and ln $\boldsymbol{log_x a \times \div b = log_x a \pm log_x b \mid b^{log_b(x)} = x}$
$\ln x^m = \ln x \qquad \log_x x^m = \log_x n \qquad (\ln x = \log_e x \,/\, e^y)$
$m \ln x = \ln x \qquad m \log_x x = \log_x n \qquad (\ln e = 1 \mid \ln 1 = 0)$

SETS / LAWS **Powerset Formula:** $\boldsymbol{P(A) = 2^n}$ (All subsets of A)
A − B = A\B | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.0 |
A ⊕ B = (A ∪ B) − (A ∩ B)
Compliments: Ā, A^c, A'
Commutative Law: $A \cup B = B \cup A \mid A \cap B = B \cap A$
Associative Law: $(A \oplus B) \oplus C = A \oplus (B \oplus C)$
De Morgan's Law: $\overline{(A \cup B)} = \bar{A} \cap \bar{B} \mid \overline{(A \cap B)} = \bar{A} \cup \bar{B}$
Distributive Law: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
Identity Law: $A \cup \emptyset = A \mid A \cap U = A$
Domination Law: $A \cup U = U \mid A \cap \emptyset = \emptyset$
Idempotent Law: $A \cap A = A \mid A \cup A = A$
Complement Law: $A \cup \bar{A} = U \mid A \cap \bar{A} = \emptyset$
$\bar{U} = \emptyset \mid \bar{\emptyset} = U$
Double Negation: $\overline{\overline{(A)}} = A$
Annihilation Law: $\mathbb{U} \cup (\bar{A} \cap B \cap C) = \mathbb{U} \mid (\mathbb{U} \cap \bar{A} \cap B) = \mathbb{U}$
$A - A = \emptyset \mid A - B = A \cap \bar{B}$
Absorption Law: $A \cup (A \cap B) = A$; $A \cap (A \cup B) = A$

Membership table/Truth Table

p	q	r	¬q	(p ∧ ¬q)	(p ∧ ¬q) → r	p ∨ r	q → r	(p ∨ r) ↔ (q → r)
T	T	T	F	F	T	T	T	T
F	F	T	T	F	T	T	T	T
T	F	T	T	T	T	T	T	T
F	T	T	F	F	T	T	T	T
T	T	F	F	F	T	T	F	F
F	F	F	T	F	T	F	T	F
T	F	F	T	T	F	T	T	T
F	T	F	F	F	T	F	F	T

A	B	(A ∪ B)	(A ∩ B)	(A ⊕ B)	Ā	B̄	$\overline{(A \cup B)}$	$\bar{A} \cap \bar{B}$
0	0	0	0	0	1	1	1	1
1	0	1	0	1	0	1	0	0
0	1	1	0	1	1	0	0	0
1	1	1	1	0	0	0	0	0

Observe from the Membership Table that
$\overline{(A \cup B)}$ and $\bar{A} \cap \bar{B}$ columns identical.

Partition of a set S (or Equivalence Classes)
non-empty subsets, mutually disjointed. (No common element)
Union of all the subsets = S.

Pick a Point Method **Prove that** $\boldsymbol{J \subseteq M}$
Prove that: $\boldsymbol{P(A) \cup P(B) \subseteq P(A \cup B)}$
Let $x \in P(A) \cup P(B) \Rightarrow x \in P(A)$ **or** $x \in P(B)$
$x \subseteq A$ **or** $x \subseteq B \;\Rightarrow\; x \subseteq A \subseteq A \cup B$ **or** $x \subseteq B \subseteq A \cup B$
$\Rightarrow x \subseteq A \cup B$ **or** $x \subseteq A \cup B$
In either case we have: $x \subseteq A \cup B, which\ x \in P(A \cup B)$
Since X is arbitrary, $P(A) \cup P(B) \subseteq P(A \cup B)$ **(Proven)**

FUNCTIONS: Range subset of Codomain $(\mathbf{R_f \subseteq C_f})$
All elements of Domain D_f have to be mapped to C_f
a = Pre-Image, b = Image $f(a) = b$
Brackets: a is included but b is not: [a,b[or [a,b)
(We do not include ∞) so] − ∞, ∞[

Vertical line test: If a vertical line cuts the graph twice it will not be a function.

Horizontal line test:
Injective (one-to-one): Any horizontal line intersects the graph at most one time.
Surjective (onto): Every horizontal line intersects the graph at least once.
Bijective: Both injective and surjective it is Bijective hence it is invertible.

Example: Prove that $\boldsymbol{f: \mathbb{R} \mapsto \mathbb{R}^+, f(x) = 2^{x+3}}$ **is bijective and Find its inverse function.**
Injective: Suppose $f(a) = f(b) \;\Rightarrow 2^{a+3} = 2^{b+3}$
$\Rightarrow \log_2 2^{a+3} = \log_2 2^{b+3} \Rightarrow a + 3 = b + 3 \Rightarrow a = b$
Therefore, f is Injective.
Surjective: Let $y \in C_f = \mathbb{R}^+$, where we find at least one Pre-Image for y. Solving for $f(x) = y$
$\Rightarrow 2^{x+3} = y \Rightarrow \log_2(2^{x+3}) = \log_2(y) \Rightarrow x + 3 = \log_2(y)$
$x = \log_2(y) - 3 \;\in \mathbb{R} = D_f$
Therefore, f is subjective, hence f is a bijective function as it is both Injective and Subjective. **(Proven)**
The Inverse function is $f^{-1}: \mathbb{R}^+ \mapsto \mathbb{R}\ where\ f^{-1}(x) = \log_2(x) - 3$ ∎

Cartesian Product
$A = \{1,2\}, B = \{3\} \quad A \times B = \{(1,3),(2,3)\}\ where\ (x \in A, y \in B)$ | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.0 |
$A \times B \neq B \times A\ \ as\ B \times A = \{(3,1),(3,2)\}$
Is $\boldsymbol{f: \mathbb{Z} \times \mathbb{Z}^+ \mapsto Q\ \ f(m,n) = \frac{m}{n}}$ **an Bijective Function?**
Counter example: Let $f(7,7)\ and\ f(9,9)$
$f(7,7) = f(9,9) = 1\ but\ (7,7) \neq (9,9)$
Therefore, f is not injective.
Surjective Prove: Let $q \in Q$ by definition of rational numbers
, $q = \frac{m}{n}\ (where\ m \in \mathbb{Z}, n \in \mathbb{Z}^+ (b > 0))$
When $b > 0$, then (m,n) $\in \mathbb{Z} \times \mathbb{Z}^+$
Therefore f is surjective as every rational number can be written with a positive denominator. However f is not Bijective as it is not Injective. **(Proven)**

Composition of Functions $\boldsymbol{(f \circ g)(x) \neq (g \circ f)(x)}$
$\boldsymbol{f(x) = x^2 - 5\ and\ g(x) = x + 2}$
$(f \circ g)(x) = f(g(x)) = (x + 2)^2 - 5$
$(g \circ f)(x) = g(f(x)) = (x^2 - 5) + 2 = x^2 - 3$
Suppose $\boldsymbol{f: A \mapsto B, g: B \mapsto C}$ **Prove that if** $\boldsymbol{g \circ f}$ **is Onto, then g is also Onto**.
Let $(g \circ f)(x) = g(f(x)) = g(B)$
$g \circ f: A \mapsto C$ We know that $g \circ f$ is Onto meaning for every $y \in C$, there exist an $a \in A$ such that $(g \circ f)(a) = y$
$\Rightarrow g(B) = g(f(a)) = y\ (where\ f(a) \in B)$
$g: B \mapsto C$ for every $y \in C$, we need to find $b \in B$ such that $g(b) = y$
Choose $b = f(a)$
Therefore g is Onto. **(Proven)**
Prove that if $\boldsymbol{g \circ f}$ **is One-to-one, then so is f**.
$g \circ f: A \mapsto C$ We know that $g \circ f$ is One-to-one, meaning:
Suppose $(g \circ f)(a) = (g \circ f)(b) \quad g(f(a)) = g(f(b)) \;\Rightarrow a = b$
Therefore f is One-to-one. **(Proven)**

English	Propositions	Sets	Boolean
AND	∧	∩	·
OR	∨	∪	+
Or, but not Both	⊕	⊕	⊕
NOT	¬p	$\bar{P}$	$\bar{p}$
Universe	T	𝕌	1
Empty	F	∅	0
If p then q	$p \longrightarrow q$	$\bar{P} \cup Q$	$\bar{p} + q$
P if and only if q	$p \longleftrightarrow q$	$(\bar{P} \cup Q) \cap (\bar{Q} \cup P)$	$(\bar{p} + q) \cdot (\bar{q} + p)$

P **only** if q: $p \longrightarrow q$
Q if p: $p \longrightarrow q$ | p if q: $q \longrightarrow p$
Q is necessary **for** p: $p \longrightarrow q$ | p is necessary **for** q: $q \longrightarrow p$
If p then q: $p \longrightarrow q$
P implies q: $p \longrightarrow q$
For p is necessary **that** q: $p \longrightarrow q$
P is sufficient for q: $p \longrightarrow q$
Normal: : $p \longrightarrow q$ | $(p \wedge q) \longrightarrow (r \wedge s)$
Contrapositive: $\neg q \longrightarrow \neg p$ | $\neg(r \wedge s) \longrightarrow \neg(p \wedge q)$
Converse: $q \longrightarrow p$ | $(r \wedge s) \longrightarrow (p \wedge q)$
Inverse: $\neg p \longrightarrow \neg q$ | $\neg(p \wedge q) \longrightarrow \neg(r \wedge s)$
Tautology: Proposition that is **Always** True: $p \vee \neg p = T$
Contradiction: Proposition that is **Always** False: $p \wedge \neg p = F$
Contingency: Proposition that is True and False, **Mixed**

Truth List
$p \longrightarrow q = \neg q \rightarrow \neg p$ | $p \longrightarrow q = \neg p \rightarrow q$
$p \longrightarrow q = F\ \ (p = T, q = F)$ | $p \longleftrightarrow q = T\ (p = q = T/F)$

Quantifier | 2026 [Xiong Hui] | https://github.com/kohxionghui | v2.0 |
$\forall x$: **All/Every/Any/Each**
$\exists x$: There exist **Some/At Least One**
$\exists!\, x$: There exists **Exactly One**
$\forall x\ (P(x) \rightarrow Q(x))$: **Everyone** in the world **Who** is a Pro athlete plays football
$\exists x\ (x + y = 1)$: There exist **At Least One** value X which $x + y = 1$

Negation of Quantifiers
$\neg(\forall x\, \exists y\, \forall z\, P(x,y,z)) = \exists x\, \forall y\, \exists z\, \neg P(x,y,z)$
$\neg(\forall x\, (P(x) \wedge Q(x))) = \exists x\, \neg(P(x) \wedge Q(x)) = \exists x\, (\neg P(x) \vee \neg Q(x))$
Is $\boldsymbol{\forall x\, (P(x) \vee Q(x)) = \forall x\, P(x) \vee \forall x\, Q(x)}$
Counter example: Let $u = \mathbb{Z}^+, P(x)$: x is even, $Q(x)$: x is odd
LHS: Every integer is either odd **or** even (True)
RHS: Every integer is even **or** Every integer is odd (False)
Therefore $\forall x\, (P(x) \vee Q(x)) \neq \forall x\, P(x) \vee \forall x\, Q(x)$ as $LHS \neq RHS$

Class 7 $\boldsymbol{(\bar{x} \cdot y \cdot \bar{z}) + \overline{(x + y)} \cdot (x + \bar{y} + z)}$
$= \overline{(\bar{x} \cdot y \cdot \bar{z})} \cdot \overline{(x + y)} \cdot (\bar{x} \cdot \bar{y} \cdot \bar{z})$ (De Morgan's Law)
$= (\bar{\bar{x}} + \bar{y} + \bar{\bar{z}}) \cdot \overline{(x + y)} \cdot (\bar{x} \cdot \bar{y} \cdot \bar{z})$ (De Morgan's Law)
$= (x + \bar{y} + z) \cdot (x + y) \cdot (\bar{x} \cdot y \cdot \bar{z})$ (Double Negation)
$= (xx + x\bar{y} + xz + yx + y\bar{y} + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$ (Distributive Law)
$= (x + x\bar{y} + xz + yx + 0 + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$
(Idempotent and Compliment Law)
$= (x + x\bar{y} + xz + yx + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$ (Identity Law)
$= ((x + yx) + x\bar{y} + xz + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$
(Commutative and Associative Law)
$= ((x) + x\bar{y} + xz + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$ (Absorption Law)
$= (x(1 + \bar{y} + z) + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$ (Distributive Law)
$= (x \cdot 1 + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$ (Annihilation Law)
$= (x + yz) \cdot (\bar{x} \cdot y \cdot \bar{z})$ (Identity Law)
$= (x\bar{x}\bar{y}\bar{z} + \bar{x}yyz\bar{z})$ (Distributive and Commutative Law)
$= (0 \cdot y\bar{z} + \bar{x}yy \cdot 0)$ (Complement Law)
$= (0 + 0)$ (Annihilation Law)
$= 0$ (Idempotent Law) ∎

Direct Prove: **Prove if n is an odd integer, then** $\mathbf{n^2}$ **is odd.**
Let n be odd: $n = 2k + 1, k \in \mathbb{Z}$
$n^2 = (2k + 1)^2 \;=\; 4k^2 + 4k + 1 \;=\; 2(2k^2 + 2k) + 1$
$\Rightarrow\ (2k^2 + 2k) \in \mathbb{Z}$
Therefore, n^2 is odd. **(Proven)**

Indirect prove: **Prove if 3n+2 is odd, then n is odd.**
Prove by contrapositive: If n is even, then 3n+2 is even.
Let n be even: $n = 2k\ (k \in \mathbb{Z})$
Multiple $n = 2k$ by 3: $3n = 6k$
$\Rightarrow 3n + 2 = (6k) + 2 \;=\; 2(3k + 1)$
$\Rightarrow\ (3k + 1) \in \mathbb{Z}$
Therefore, 3n+2 is even. **(Proven)**

Fundamental Theorem of Arithmetic
Given Any Integer $n > 1$, n can always be written as a Product of Primes **OR** a Prime itself.

Prove there are infinitely many primes.
Assume finite number of primes:
Let $P = \{p_1, p_2, p_3, \ldots, p_k\}$ where p_k is the largest prime.
Consider a new integer $N \;=\; (p_1 \times p_2 \times p_3 \times \ldots \times p_k) \;+\; 1$
By the Fundamental Theorem of Arithmetic, either N is a Prime or a Product of Primes.
However, N cannot be a prime as $N > P = \{p_1, p_2, p_3, \ldots, p_k\}$
If N is divided by Any Primes in Set P, there is always a remainder of 1. Hence, N cannot be a Product of Primes. Therefore, N contradicts the Fundamental Theorem of Arithmetic.
Therefore, the number of Primes must be infinite. **(Proven)**

Rules of Inference (Existential Instantiation / Generalisation)
Modus Ponens: $p \rightarrow q,\ p \quad TF \quad q$
Modus Tollens: $p \rightarrow q,\ \neg q \quad TF \quad \neg p$
Conjunction: $p, \quad q \quad TF \quad p \wedge q$
Simplification: $p \wedge q \quad TF \quad p$
Addition: $p \quad TF \quad p \vee q$
Hypothetical Syllogism: $p \rightarrow q, \quad q \rightarrow r \quad TF \quad p \rightarrow r$
Disjunctive Syllogism: $p \vee q, \quad \neg p \quad TF \quad q$
Resolution: $p \vee q, \quad \neg p \vee r \quad TF \quad q \vee r$

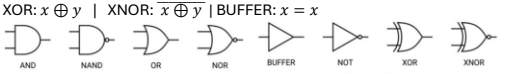
Steps	Statements	Reasons
1	$\forall x\ (S(x) \rightarrow (H(x) \rightarrow P(x)))$	Premise 1
2	$S(a) \rightarrow (H(a) \rightarrow P(a))$	Universal Instantiation of (1)
3	$\forall x\ (S(x) \rightarrow H(x))$	Premise 3
4	$S(a) \rightarrow H(a)$	Universal Instantiation of (3)
5	$S(a)$	Assumption for conditional proof of $S(a) \rightarrow P(a)$
6	$H(a) \rightarrow P(a)$	Modus Ponens using (2) and (5)
7	$H(a)$	Modus Ponens using (4) and (5)
8	$P(a)$	Modus Ponens using (6) and (7)
9	$S(a) \rightarrow P(a)$	Discharge of Assumption of (5) and (8)
10	$\forall x\ (S(x) \rightarrow P(x))$	Universal Generalisation using (9) since (a) is arbitrary

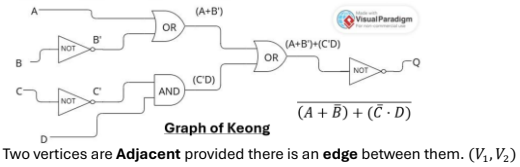
K-Map (Do after knowing the full Turth Table)

AB \ CD	00	01	11	10
00	0	0	1	1
01	0	1	0	0
11	0	0	0	0
10	1	1	0	1

(iii) $F = \bar{A} \cdot \bar{B} \cdot C + \bar{A} \cdot B \cdot \bar{C} \cdot D + A \cdot \bar{B} \cdot \bar{C} + \bar{B} \cdot C \cdot \bar{D}$ ∎
Only group 1s that are adjacent and wrap around
As long as they are Binary 1,2,4,8 groups of 1s.
Make sure all 1s are wrapped and with the lowest number of groups

Logic Gates: OR: $x + y$ | AND: $x \cdot y$ | NOT: $\bar{x}$ | NOR: $\overline{x + y}$ | NAND: $\overline{x \cdot y}$
XOR: $x \oplus y$ | XNOR: $\overline{x \oplus y}$ | BUFFER: $x = x$



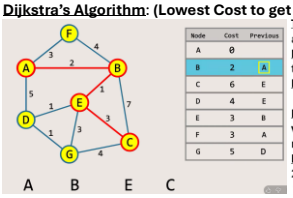


$\overline{(A + \bar{B}) + (\bar{C} \cdot D)}$

Graph of Keong

Two vertices are **Adjacent** provided there is an **edge** between them. (V_1, V_2)
Two edges are **Adjacent** if they share a common **endpoint** (Same vertex V_2).
There can be **Parallel** Edges and **Loop** Edges
An edge is **Incident** with/on a vertex if that vertex is an Endpoint of that edge
Isolated Vertex: No edges are **Incident** with that vertex. $(V_4,\ V_5)$
Degree of a vertex: Number of edges **Incident** (Connected) with that vertex
Degree Sequence of G: $\boldsymbol{V_1(1), V_2(2), V_3(1), V_4(0), V_5(0)}$
Walk: Sequence of vertices and edges where vertices and edges can repeat
Present in this way: $(V_1 e_1 V_2 e_2 V_3 e_2 V_2)$
Trail: **Walk** in which no edges can be repeated (But Vertices can repeat)
Present in this way: $e_1,\ e_2$
Circuit: Round trip, a **Closed Trail**, so the vertex your begin is also the end
Present in this way (No edges can repeat): $(V_1 e_1 V_2 e_2 V_3 e_3 V_1)$ (If there is a e_3)
Path: Sequence of vertices and edges where no vertices or edges is repeat
Present in this way: $(V_1 e_1 V_2 e_2 V_3)$
Cycle: **Closed Path** where the only vertex to be repeated is begin and end
Present in this way (No edges can repeat): $(V_1 e_1 V_2 e_2 V_3 e_3 V_1)$ (If there is a e_3)
Cycle Length: Just count the number of **Edges**
Euler Path: Where you use **All Edges Exactly Once** (Only Vertex can repeat)
Euler Circuit: A Closed Euler Path
Euler circuit and Path require all connected vertices have even degrees.
Euler path exists if exactly two vertices have odd degrees.
If more than two vertices have odd degrees, neither exists.
Hamilton Path: **Path** that visits all **Vertex Exactly Once** (Edges can repeat)
Hamilton Cycle: **Closed Hamilton Path** where the only vertex to be repeated is the one you begin and end with.
Handshaking Problem: Just count the number of **Edges**
No Multiple Handshaking / No self-handshaking / Everyone Must be Shaked
Simple Graph: No Loop edges, No Parallel edges
Complete Graph $(\boldsymbol{K_n})$: **Simple Graph** where all **vertices** are adjacent to each other / Degree of $\boldsymbol{K_n = (n-1)}$ / $\boldsymbol{K_1}$**exist as 1 vertex with no edge**
Regular Graph: Every vertex have the same degree (n-Regular)
The Handshaking Theorem: (Works for All Graphs)
Total number of Edges $= 2\,|E| = n(n-1)$ (Where n is vertices)
Graph with degree sequence 4,3,3,1
Using The Handshaking Theorem: $4 + 3 + 3 + 1 = 2\,|E|$
$2\,|E| = 11 \quad / \quad |E| = 5.5$ (Cannot have Half an Edge)
Therefore, there is no such Graph as $|E| = 5.5 \notin \mathbb{Z}^+$
For Simple Graph: $(n-1)$ is the Maximum for Degree, so a degree of 4 is not possible for a simple graph with degree sequence of 4,3,2,1.

Dijkstra's Algorithm: **(Lowest Cost to get there)**



Node	Cost	Previous
A	0	
B	2	A
C	6	E
D	4	E
E	3	B
F	3	A
G	5	D

Tree: **Simple Graph** which is **Connected** and **Acyclic (Not closed cycle)**
Minimum Spanning Tree: **Lowest Cost** to construct, keep lowest number
Kruskal's Algorithm: **(MST)** starting with **n vertices** and $(n-1)$ edges
Prim's Algorithm: **(MST)** starts with **one vertex** and adding the cheapest edge to reaches any new "unvisited" vertex
Bipartite Graphs: (Compatibility Graph) 2 sets, and edges connect only between the 2 sets and not within own set

Matching Graph: No **Adjacent** edges, 2 Vertices 1 Edge only
Hopcroft Karp Algorithm: To get Maximum Matching, when all vertices join
Digraph: $xRy\ means\ x > y$ or $x = y \mid R = \{(2,1)\}\ means\ 2R1 \mid 1 \leftarrow 2$
Relation R have these properties (All 3 = Equivalence Rational)
Reflexive – aRa for $\forall a \in A$ | Symmetric – If aRb, then bRa for $\forall a, b \in A$
Transitive – If aRb and if bRc, then aRc for $\forall a, b, c \in A$
Let $\boldsymbol{A = \{Set\ of\ all\ People\ in\ the\ world\}}$ **xRy (x is same gender as y)**
For $\forall a \in A, aRa$ because each person is same gender as himself/herself.
Therefore, R is Reflexive.
For $\forall a, b \in A$, Suppose aRb (a is the same gender as b) then bRa (b is the same gender as a as well) Therefore, R is Symmetric.
Anti-Symmetric: Suppose aRb and suppose bRa, then $a = b$ for $\forall a, b \in A$
Suppose Set B is a subset of Set A and Set A is also a subset of Set B. Then it is **anti-symmetric** as they are equal.
(Reflexive, Anti-symmetric, Transitive = Partial Order)
Comparable Elements: For $\forall a, b \in A$ Either aRb **or** bRa
(Partial Order, Comparable = Total Order)
Pigeonhole: If N objects are placed in K boxes (Categories), then at least one box contains at least $\left\lceil \frac{N}{K} \right\rceil$ $(Ceiling\ Function)$ objects.
$nCr = \frac{n!}{(n-r)!r!}$ | $nPr = \frac{n!}{(n-r)!}$ | $0! = 1$ | **General Term**: $T_{r+1} = nCr\ a^{n-r} b^r$
Binomial Theorem: $(a + b)^n = a^n + nC1\ a^{n-1} b^1 + nC2\ a^{n-2} b^2 + .. + b^n$
Pascal's Identity: $nC0 + nC1 + nC2 + .. + nCn = 2^n$ | **Probability**: $\frac{n!}{c!p!q!}$